

SOLUTION ARCHITECTURE DOCUMENT

AI Powered Style Trained Social Media Post Generator

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Status	Draft For Engineering Review
Based on	SRS v1.0 AI Post Generator Requirements

1. Project Goal

This document defines the technical architecture that satisfies every requirement stated in the Software Requirements Specification (SRS v1.0). The system must enable the marketing team to produce social media posts that achieve the following outcomes as defined in the SRS:

SRS Outcome	What the Architecture Must Deliver
Authentic voice	Posts written in the exact voice of a trained persona (Zack, Luke, etc.), indistinguishable from that writer's real posts by a human reviewer familiar with the writer.
Zero fabricated content	Every fact, statistic, and claim in the output is sourced exclusively from the user supplied brief. Nothing is invented.
Platform aware output	The same brief produces structurally different posts for X (Twitter) and Facebook, with platform constraints applied automatically.
Non-technical operation	Any marketing team member can operate the system with no engineering knowledge or assistance.
Scalable persona management	New writer personas are added by ingesting training data only. No code changes are required.

2. Problem Statement

The SRS identifies five specific problems with current general purpose AI tools that this architecture must eliminate:

#	Problem (from SRS)	Architecture Requirement
1	Generic, recognisably AI generated wording	Style engine trained on each writer's real posts must produce output that sounds like the human, not like a generic AI.
2	Inconsistent voice posts do not match the intended writer	Each persona has an isolated training dataset. The system must not mix styles across personas.
3	Platform mismatch posts formatted incorrectly for the target platform	Platform constraints for X (Twitter) and Facebook must be applied automatically based on the user's selection.
4	Low reach as a direct result of the above problems	Authentic voice + correct platform format directly addresses the root causes of low reach.

3. Scope

The architecture covers the following scope as defined in SRS Section 3:

- 4 named writer personas (Zack, Luke, Alex, Writer 4 to be confirmed by marketing team before development).
- 2 target platforms: X (Twitter) and Facebook.
- User supplied brief as the sole factual source for all generated content.
- First release only. Additional personas are added via data ingestion, requiring no architectural changes.

3. Two Proposed Solutions

Solution1: JSON Persona + AI API (Fast Prototype)

How it delivers the required outcomes

Each writer persona is stored as a JSON configuration file containing a tone description, style rules, and a curated set of example posts (drawn from the training dataset). When the user submits a brief, the backend combines the persona JSON with the user's context and the target platform constraints, then sends the assembled prompt to an AI API in a single call. The model returns a post styled according to the persona definition.

End to End Request Flow

Step	Stage	What Happens
1	User submits brief	Marketing team member provides topic, key facts, selected persona (Zack / Luke / Writer 3 / Writer 4), and target platform (X or Facebook).
2	Persona JSON loaded	Backend reads the JSON file for the selected writer. File contains tone descriptor, style rules, and curated example posts.
3	Platform constraints applied	Backend appends platform specific rules: X character limit and short form structure, or Facebook long form structure. Applied automatically user does not configure this.
4	Prompt constructed	A single prompt is assembled: persona JSON + platform constraints + user's brief. The brief is presented as the only factual source.
5	AI API call	Assembled prompt is sent to the AI API as a single completion request.
6	Post returned to user	The generated post is returned, labelled with the writer's name and target platform, ready for review and publishing.

SRS Outcomes Addressed

SRS Outcome	Addressed?	Notes
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Authentic writer voice	Partially	Style fidelity depends on quality of JSON curation. Output may not always be indistinguishable from the real writer.
Zero fabricated content	Yes	The prompt explicitly restricts the model to user supplied facts only. No external data is introduced.
Platform aware output	Yes	Platform constraints are injected into the prompt automatically.
Non technical operation	Yes	Interface presents a simple form: brief, persona selector, platform selector.
Scalable persona management	Yes	Adding a new persona requires only creating a new JSON file. No code changes.

Best for: MVP · Proof of concept · Fast first delivery · Persona validation



Solution 2 RAG (Retrieval Augmented Generation) ★ Best Overall

How it delivers the required outcomes

All training posts for each writer are converted into vector embeddings and stored in a vector database, with one isolated collection per persona. When a user submits a brief, the system converts the brief into an embedding and retrieves the 3 to 5 most stylistically similar real posts from that writer's collection. Those retrieved examples combined with the user's brief and the platform constraints are sent to the AI API to generate the post. The model now has real, contextually relevant examples of the writer's actual style, not a static text description.

End to End Request Flow

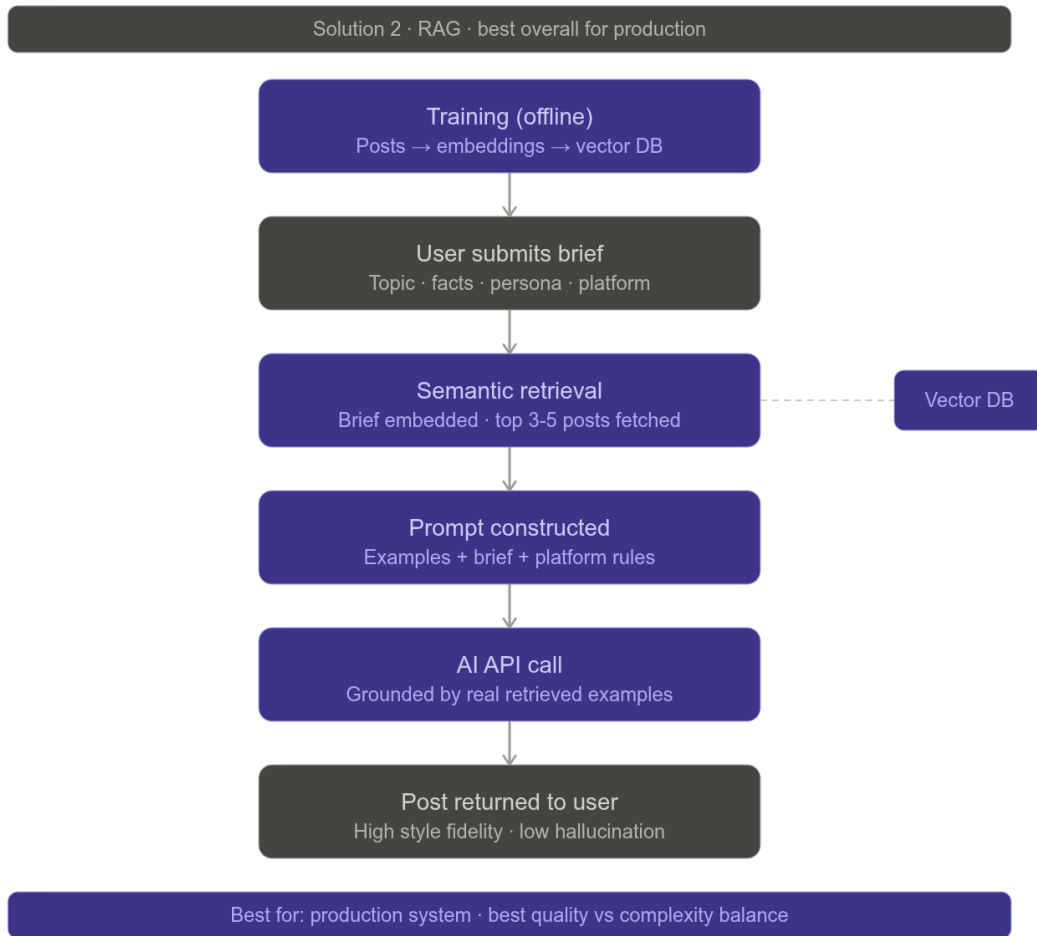
Step	Stage	What Happens
1	Training (offline)	Marketing team supplies curated posts (minimum 50 per persona, per SRS Section 4.2). The ingestion pipeline converts each post into a vector embedding and stores it in the writer's isolated collection in the vector database.
2	User submits brief	Marketing team member provides topic, key facts, selected persona (Zack / Luke / Writer 3 / Writer 4), and target platform (X or Facebook).
3	Semantic retrieval	The brief is converted into an embedding. The system queries the selected writer's vector collection and retrieves the 3 to 5 posts whose style is most similar to the brief's topic and tone.
4	Platform constraints applied	Platform specific rules are loaded automatically: X short form character limit, or Facebook long form structure. Applied without user configuration.
5	Prompt constructed	A prompt is assembled from: retrieved real example posts + platform constraints + user's brief. The brief is presented as the only factual source.
6	AI API call	The assembled prompt is sent to the AI API. The model generates a post grounded in the writer's real examples, not a generalised style description.

7	Post returned to user	The generated post is returned, labelled with the writer's name and target platform, ready for review and publishing. If all styles mode was selected, all four outputs are shown side by side.
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SRS Outcomes Addressed

SRS Outcome	Addressed?	Notes
Authentic writer voice	Yes High fidelity	The model generates from real retrieved examples of the writer's posts, not a text description. Style fidelity is significantly higher than Solution 1 and meets the SRS requirement of being indistinguishable from authentic posts by a human reviewer familiar with the writer.
Zero fabricated content	Yes	The prompt explicitly restricts the model to user supplied facts only. Retrieved examples are style references, not factual sources.
Platform aware output	Yes	Platform constraints are applied automatically at prompt construction time. Same brief produces structurally different outputs for X and Facebook.
Non technical operation	Yes	Interface is identical to Solution 1: brief, persona selector, platform selector. The retrieval layer is invisible to the user.
Scalable persona management	Yes Fully scalable	Adding a new persona requires running the ingestion pipeline on the new writer's posts. No code changes. Quality improves automatically as more posts are added.

Best for: Production system · Full SRS compliance · Best balance of quality and build effort



6. Side by Side Comparison

Criterion	Solution 1 JSON + API	Solution 2 RAG
Build complexity	Low (days)	Medium (weeks)
Style fidelity vs SRS requirement	Moderate may not meet indistinguishability requirement	High meets SRS indistinguishability requirement
Zero fabricated content	Yes	Yes
Platform aware output	Yes	Yes
Scalability (new personas)	Yes JSON file only	Yes data ingestion only

Non technical operation	Yes	Yes
All styles comparison output	Yes	Yes
Recommended for	MVP / Prototype	Production system

7. Recommendation

If the priority is...	Recommended solution
Speed validate personas and collect marketing team feedback fast	Solution 1: JSON + API. Deliverable in days. Use this to confirm the four writer personas and collect feedback before committing to the full build.
Full SRS compliance meeting the indistinguishability and scalability requirements in production	Solution 2: RAG. This is the recommended architecture for the production system. It is the only solution that fully satisfies all SRS outcomes, particularly style fidelity and training dataset integration.

Recommended path: Start with Solution 1 to validate personas with the marketing team. Once the four writer personas and their training datasets are confirmed and curated (minimum 50 posts per persona, as required by SRS Section 4.2), migrate to Solution 2 for the production release.